

Qatar-1b

R-2121

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_121101 · created 2026-08-11T01:10:11+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456232.68325 +/- 0.0001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.017
Transit depth (Rp/Rs)^2	2.88 +/- 0.57 [%]
Orbital Inclination (inc)	84.07 +/- 1.23
Airmass coefficient 1 (a1)	0.952 +/- 0.019
Airmass coefficient 2 (a2)	0.035 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.96 %
Best Comparison Star	#2 - [313, 268]
Optimal Method	PSF photometry
Transit Duration (day)	0.0703 +/- 0.0069

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.17 ± 0.017 vs 0.1463 (deviation 1.39σ)
Duration fit vs published	0.0703 d vs 0.0692 d (deviation 0.16σ)
Mid-time O–C	0.2 min
Depth SNR	10.0
Residual scatter	1.96 %

Light curve

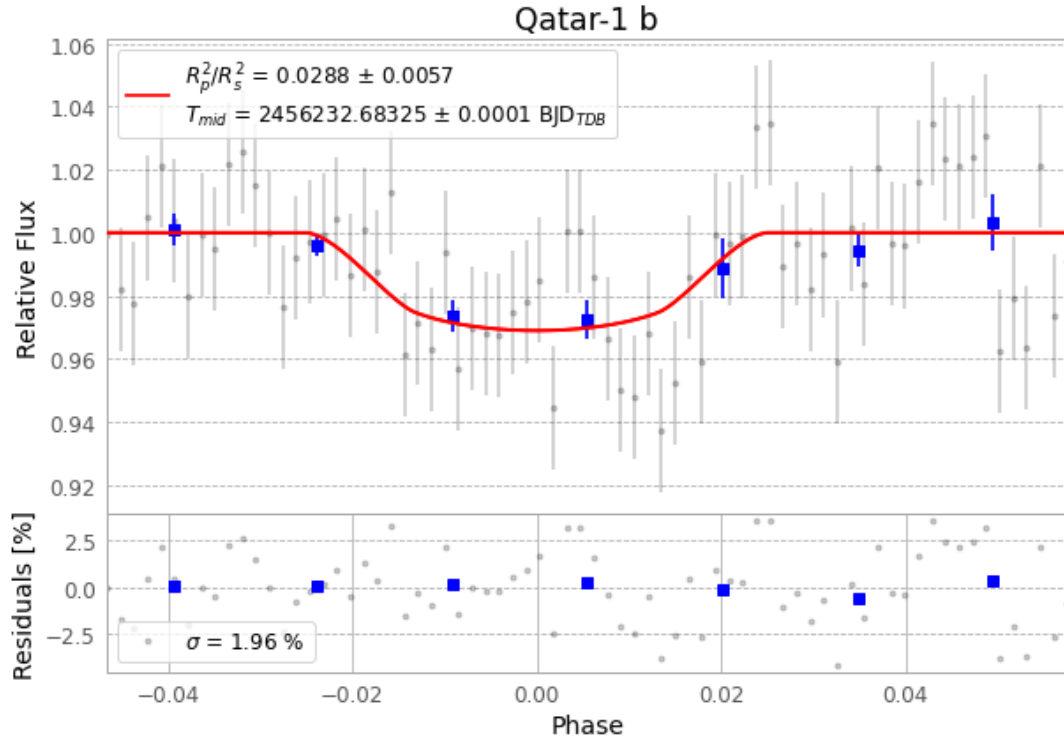


Figure 1 — FinalLightCurve_Qatar-1 b_2012-10-31.png (SHA-256 db1d3c6bfe184d4d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [382, 246], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dcf32892b25ccbb1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

72 FITS frames (47 MB), Qatar-1121101024513.FITS → Qatar-1121101061812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-121101.log (87b459e16979...), FinalLightCurve_Qatar-1 b_2012-10-31.png (db1d3c6bfe18...), FinalLightCurve_Qatar-1 b_2012-10-31.csv (da6006929a77...)

Compute resources

Wall time 187.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 01:10Z.